1a) Cutting ratio:

$$r = \frac{t_0}{t_c} = \frac{0.15}{0.2} = 0.75$$

$$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha} = \frac{0.75 \times \cos 15^\circ}{1-(0.75\sin 15^\circ)}$$

$$= 0.898941816 7$$

$$\approx 0.90$$

$$\phi = 41.95369794^\circ$$

$$\approx 42^\circ$$

b) $R = \sqrt{F_c^2 + F_t^2}$

$$= \sqrt{500^2 + 200^2} = 538.5164807$$

$$\approx 538.5 N$$

$$F_c = R\cos(\beta - \alpha)$$

$$500 = 538.5\cos(\beta - 15)$$

$$\beta = 36.80140949$$

$$\approx 36.8^\circ$$

1c) Power (cutting) $= F_c v$

Power (shearing) $= F_s v_s$

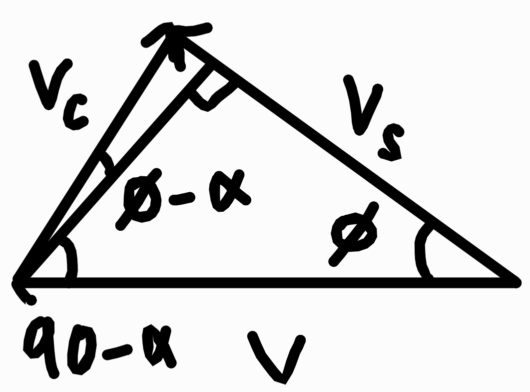
Power (friction) $= F v_c$

Percentage energy dissipated in the shear plane $= \dfrac{F_s v_s}{F_c v_c}$

From the force diagram:

$$F_s = R\cos(\phi + \beta - \alpha)$$
$$= 538.5\cos(42 + 36.8 - 15)$$
$$= 238.1366944$$
$$\approx 238.1\text{N}$$

From the velocity diagram:



$$\frac{v}{\sin(90 - \phi + \alpha)} = \frac{v_s}{\sin(90 - \alpha)}$$

$$\frac{v}{\cos(\phi - \alpha)} = \frac{v_s}{\cos\alpha}$$

$$v_s = \frac{\cos\alpha}{\cos(\phi - \alpha)} v = \frac{\cos 15}{\cos(42 - 15)} \times 2$$
$$= 2.167276108$$
$$\approx 2.16\text{ms}^{-1}$$

$$\frac{F_s v_s}{F_c v_c} = \frac{238.1(2.16)}{500 \times 2}$$
$$= 0.5151079683$$
$$\approx 52\%$$

2) Specific energy is the amount of energy required to remove a unit volume of work material ($J\ mm^{-3}$). The specific energy is usually larger for stronger work materials.

Size effect refers to the relationship between the specific energy and chip size. As specific energy increases, the chip size decreases.
Micro-machining tends to generate small chip sizes. Hence, the process is expected to consume more energy.

3) $V = \pi D_{avg} N$

$$N = \frac{V}{\pi D_{avg}} = \frac{180}{\pi(125\times10^{-3})} = 458.3662361$$
$$\approx 458.3 \text{ rev min}^{-1}$$

$$t = \frac{1}{fN} = \frac{300}{0.225\times458} = 2.908882087 \text{ min}$$
$$\approx 2.9 \text{ min}$$

$$T = 5 \times 2.9 = 14.5444043 \text{ min}$$
$$= 14.54 \text{ min}$$

Taylor's tool life equation:

$$VT^n = C$$
$$180(14.54)^n = C \quad -(1)$$

$V = 120$

$$N = \frac{120}{\pi(125\times10^{-3})} = 305.5774907$$
$$\approx 305.6 \text{ rev min}^{-1}$$

$$t = \frac{1}{fN} = \frac{300}{305.6\times0.225} = \frac{25}{18}\pi \approx 4.4 \text{ min}$$

$$T = 25\times4.4 = \frac{625}{18}\pi \approx 109.1 \text{ min}$$
$$120\left(\frac{625}{18}\pi\right)^n = C \quad -(2)$$

From (1):

$$180(14.54)^n = C$$
$$n\log 14.54 = \log\left(\frac{C}{180}\right)$$
$$\log C - n\log 14.54 = \log 180$$

From (2):

$$120\left(\frac{625}{18}\pi\right)^n = C$$
$$n\log\left(\frac{625}{18}\pi\right) = \log\left(\frac{C}{120}\right)$$
$$\log C - n\log\left(\frac{625}{18}\pi\right) = \log 120$$

Solving:

$\log C = 2.489245408$ $\quad\therefore C = 308.4930668 \approx 308.5$

$n = 0.2012330638 \approx 0.2$ $\quad\therefore VT^{0.2} = 308.5$

4) Material removal rate:

$$MRR = dfV$$

$$d = \frac{D_o - D_f}{2} = \frac{75-65}{2} = 5\,mm$$

$$fN = 250\,mm\ min^{-1}$$

$$f = \frac{250}{400} = 0.625\,mm\ rev^{-1}$$

$$V = \pi D_{avg} N = \pi\left(\frac{75+65}{2}\right)\times\left(\frac{400}{60}\right)$$

$$= \frac{1400}{3}\pi$$

$$MRR = dfV$$

$$= 5(0.625)\left(\frac{1400}{3}\pi\right)$$

$$= \frac{4375}{3}\pi$$

$$\approx 4581\,mm^3\,s^{-1}$$

Machining time:

$$t_m = \frac{l}{fN} = \frac{200}{250} = 0.8\ min$$

Power:

$$u_t = \frac{Power}{MRR}$$

$$Power = u_t \times MRR$$

$$= 3.5 \times \frac{4375}{3}\pi$$

$$= \frac{30625}{6}\pi\ W$$

$$\approx 16\,kW$$

5) Finish diameter:

$$d = \frac{D_o - D_f}{2}$$

$$12 = \frac{100 - D_f}{2}$$

$$D_f = 76\,mm$$

Machining time:

$$Power = 0.7 \times 3$$
$$= 2.1\,kW$$

$$u_t = \frac{Power}{MRR}$$

$$MRR = \frac{2.1 \times 10^3}{2.3}$$
$$= \frac{21000}{23} \approx 913\,mm^3\,s^{-1}$$

$$MRR = \frac{\text{Volume cut}}{\text{Time taken}} = \frac{\frac{\pi(D_0^2 - D_f^2)}{4} \times 50}{t_m}$$

$$913 t_m = \frac{50\pi(100^2 - 76^2)}{4}$$

$$t_m = \frac{2024}{35}\pi$$
$$\approx 182\,s$$

For finishing cut:

$$t_m = \frac{l}{fN} \qquad V = \pi D_{avg} N$$

$$N = \frac{V}{\pi D_{avg}} = \frac{90 \times 10^3}{\pi \times 73} = 392.436846$$
$$\approx 392.4\ \text{rev min}^{-1}$$

$$t_m = \frac{50}{0.1(392.4)} = \frac{73}{180}\pi \approx 1.27\,min$$
$$\approx 76.4\,s$$

5) Surface roughness:

$$R_a = \frac{f^2}{32R} = \frac{(0.1)^2}{32 \times 0.8} = \frac{1}{2560} \approx 0.39\,\mu m$$

$$R_t = \frac{f^2}{8R} = \frac{(0.1)^2}{8 \times 0.8} = \frac{1}{640} \approx 1.56\,\mu m$$

6) Material removal rate:

$$MRR = \frac{\pi D^2}{4} f N$$

$$= \frac{\pi (12)^2}{4} \times 0.25 \times 5$$

$$= 45\pi \approx 141.4\,mm^3\,s^{-1}$$

Drilling time:

Length drilled $= 20 + A$

$$A = \frac{1}{2} D \tan\left(90 - \frac{\theta}{2}\right)$$

$$= \frac{1}{2}(12) \tan\left(90 - \frac{118}{2}\right)$$

$$= 3.605163714$$

$$\approx 3.6\,mm$$

$$t_m = \frac{2+A}{fN} = \frac{20+3.6}{0.25 \times 5} = 18.88413097$$

$$\approx 18.9\,s$$

Torque on the drill:

$$\text{Power} = U_t \times MRR = \tau \times \omega$$

$$\tau = \frac{U_t \times MRR}{\omega}$$

$N = 5$ rev s^{-1}

$$\omega = 2\pi N = 2\pi(5) = 10\pi \text{ rads}^{-1}$$

$$\tau = \frac{2 \times 45\pi}{10\pi} = 9\,Nm$$